Bailey Wu

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RELEVANT EXPERIENCE

Senior Materials & Process Engineer **August 2022 – May 2024**
Raytheon Technologies - Raytheon Vision Systems ***Goleta, CA***

- Oversaw cross-functional engineering teams supporting high-mix, low-volume programs totaling $300M+ across MWIR, SWIR, and visible-spectrum camera systems for mission-critical space applications.
- Applied Statistical Process Control (SPC), Cp/Cpk analysis, and process capability studies to key performance indicators to reduce variation, improve yield, and ensure compliance with aerospace regulation standards.
- Conducted advanced material testing and characterization including metallurgical analysis, thermal analysis, hardness testing, and microstructure evaluation to validate material and joint integrity of weldments.
- Directed procurement and qualification of $1M+ in capital equipment by overseeing technical specifications, process validation, Gauge R&R studies, and measurable improvements in manufacturing capability.
- Trained and mentored engineers and technicians in space manufacturing best practices, hardware handling, contamination control, and process documentation standards.

Semiconductor Process Engineer **March 2021 – August 2022**
Lockheed Martin – Missiles Fire Control ***Goleta, CA***

- Engineered automated assembly lines integrating die bonders, wirebonders, and precision alignment systems, developing control logic, process interlocks, and equipment interfaces to maximize repeatability and ensure robust process control for high-reliability semiconductor hardware.
- Developed structured V&V frameworks encompassing non-destructive evaluation, vibration analysis, and thermal cycling to validate process and hardware performance under field environmental conditions.
- Led failure analysis investigations using cross-sectioning and scanning electron microscopy (SEM), applying systematic root cause analysis and CAPA methodology to improve long-term reliability of assemblies.
- Presented Production Readiness Reviews (PRR), Critical Design Reviews (CDR), and Table Top Reviews (TTR) to multi-level defense contractor and government stakeholders.

ADDITIONAL ENGINEERING EXPERIENCE

Senior Tooling Design Engineer **June 2024 – Current**
The Boeing Company – Boeing Commercial Airplanes ***Seattle, WA***

- Responsible for simulation-driven mechanical design of complex production tooling and structural systems, performing FEA in ANSYS across static, dynamic, and thermal loading conditions to support validation and production.
- Applied stress, fatigue, and tolerance stack-up analysis to evaluate structural integrity and manufacturability of production tooling against critical aerospace safety and performance requirements.
- Developed detailed 3D models, assemblies, and manufacturing drawings in CATIA applying DFM principles while maintaining engineering data integrity through PLM systems with full version control and configuration traceability.

Manufacturing Engineer **July 2018 – March 2021**
Raytheon Technologies – Raytheon Vision Systems ***Goleta, CA***

- Directed hands-on assembly operations within a Class 7, Grade C, Level 3 cleanroom environment, ensuring full compliance with space manufacturing and contamination control standards.
- Applied Lean and Six Sigma methodologies to root cause corrective actions, driving process improvements that reduced rework, enhanced compliance, and strengthened production readiness across high-reliability programs.
- Managed programs using Earned Value Management (EVMS), developing and controlling engineering budgets while overseeing process development for CRAD products transitioning into full scale production.

EDUCATION

UNIVERSITY OF WASHINGTON **June 2025**
Master of Science in Mechanical Engineering ***Seattle, WA***

UNIVERSITY OF CALIFORNIA, SANTA BARBARA **June 2018**
Bachelor of Science in Mechanical Engineering ***Santa Barbara, CA***

SKILLS

CATIA V5, SolidWorks, PTC Creo, ENOVIA, ANSYS, MATLAB, Python, Minitab, JMP, GD&T, PFMEA, EVMS